PAU SABATER NÁCHER

Valencian based in Boston, MA Updated on October 21, 2026 +1 857 639 0510
pau.sabnac@gmail.com github.com/pausabater13/ linkedin.com/in/pau-sabater/ +34 618567575

EDUCATION

Northeastern University — Boston, MA, USA Sep2025 - Current
Visiting Scholar in Applied Physics and Mathematics. THz sensing and wavefront engineering for NLOS wireless communications using Bessel (Mathieu) and Airy (Webber) beams, and validation through advanced testbeds and simulators including TeraNova and Colosseum.

CFIS, Interdisciplinary Higher Education Centre — Barcelona, Catalunya, Spain Sept2021 - 2025
Universitat Politècnica de Catalunya (UPC), BarcelonaTech
Awarded 5-year tuition fee waiver and accommodation scholarship (based on academic merit). Selective honors program for high-achieving students focused on interdisciplinary studies in mathematics, engineering and science.

- Bachelor of Science in Mathematics. Main Academic Interests:
  Combinatorics. Finance & Stochastic Calculus. Quantum Computing. Structs & Algorithms.
- Bachelor of Engineering in Physics. Main Academic Interests:
  Control Theory. Physic Electronics & EM waves. Signal Theory. Thermodynamics.

Maristas Mediterranea — Algemesí, Comunitat Valenciana, Spain *2019-2021*
High-school diploma, with honors. University admission mark: 13.61/14

RESEARCH INTERNSHIPS

Researcher — Ultrabroadband Nanonetworking (UN) Laboratory, Boston, MA, USA Sep2025 - Current

· Advanced THz-based wavefront engineering for robust long-range wireless 6G communication in obstacle-rich environments. Mathematical exploration of the use of Bessel and Airy beams for non-line-of-sight (NLOS) scenarios, targeting improved beam stability and energy efficiency at THz frequencies. Proposing a change of telecommunications paradigm to near-field, non-paraxial. Validation using Northeastern University's testbeds, simulation platforms, and Colosseum for system-level performance evaluation.

Data Scientist — REVER (Y Combinator S22) Jun2025 - Current

· Designed and deployed predictive and recommendation models for reverse e-commerce logistics, directly supporting decision-making across operations and strategy. Contributed to systems handling approximately $150M in weekly Gross Transaction Value (GTV), ensuring scalability, traceability, and performance of ML solutions in production environments.

AI-Driven Systems Intern — i2CAT Foundation / NEC laboratories Jun2024 - Sept2024

· Intelligent Systems and 6G Communications Researcher with expertise in simulation models developement for THF band systems, applying AI techniques (Transformer architectures) to analyze and recognize Emotion patterns in 3D PointCloud data. Designed high-frequency hardware. This work contributed to EU deliverables and 2 upcoming publications.

ML Engineer Intern — Esperanto Technologies Jun2023 - Sept2023

· Developed and optimized massively parallel AI accelerators for ML applications (image classifiers, object detection, recommendation systems, Diffusion TtI models & NLP programs), integrating RISC-V (ETSOC-1 devices) tensorization & vectorization operators, and inference servers with CI/CD pipelines. Proficient in PyTorch, TensorFlow, ONNX, and advanced scalable computational frameworks for data-driven optimization.

Research Intern — CSIC-UPC Institut Robòtica i Informàtica Industrial, Barcelona Feb2023-Jun2023

· Banco Santander + UPC IniRec grant recipient. Research in the field of Control theory. Developed and optimized solutions for drone thrust control systems performance, focusing on high-dynamics procedures and communication protocols (OneShot125) for precise motor control. STM32 Cube/MX/Prog/CubeIDE Microcontroller programming and experience with QGroundControl.

Electronic Researcher — UPC EcoRacing FS team, ESEIAAT Sept2021-Mar2022

· DC-AC inverter design planning so as to substitute sponsor's one. PCB design (Proteus) and micro-controller programming for legal car improvement regulations under competition requirements updates.
Technical and economic reports & Team logistics management for international competitions.

ACHIEVEMENTS, AWARDS AND PRIZES

Realfooder HACKATHON - 3rd place *February 2022*
IMMUNE Tech Institute, Madrid. JavaScript database management for food distribution optimization.
CFIS 5 years accommodation scholarship for double degree studies. *Barcelona, July 2021*
Medalla Sos - Best academic trajectory in Maristas School *June 2021*
Guadalaviar Mathematics Olympiad - 3rd place *April 2021*
Valencia mathematics Olympiad. Sponsored by Banco Santander, and U-Tad Tech Institute.
BIOLOGY National Olympiad OEB - Comunitat Valenciana Awards *March 2021*
Theoretical and practical skills competition.
Generalitat Valenciana Academic Achievements Awards *Sept 2019*
Recognition for excellent academic record. Average mark: 9.98/10
Al-kwarismi Mathematics Olympiad - Comunitat Valenciana 1st place *May 2019*
Real Sociedad Española de Matemáticas Olympiad for last high-school courses.
Puig Adam Mathematics Olympiad - 2nd place *March 2019*
Universidad Complutense de Madrid mathematic olympiad for last high-school courses.
Kangaroo Mathematic Olympiad - Comunitat Valenciana 6th place *March 2019*
Guadalaviar Mathematics Olympiad - 1st place & Special Mention *April 2018*
Valencia mathematics olympiad. Sponsored by Banco Santander, and U-Tad Tech Institute.
Kangaroo Mathematic Olympiad - Comunitat Valenciana 5th place *March 2018*
Al-kwarismi Mathematics Olympiad - Comunitat Valenciana 7th place *May 2017*
Real Sociedad Española de Matemáticas olympiad for former high-school courses.
Guadalaviar Mathematics Olympiad - 1st place & Special Mention *April 2016*
Valencia mathematics olympiad. Sponsored by Banco Santander, and U-Tad Tech Institute.
Generalitat Valenciana Academic Achivements Awards *Sept 2015*
Recognition for excellent academic record.
Al-kwarismi Mathematics Olympiad - Comunitat Valenciana 1st place *May 2015*
Real Sociedad Española de Matemáticas olympiad for children.
Guadalaviar Mathematics Olympiad - 1st place *April 2014*
Valencia mathematics olympiad. Sponsored by Banco Santander, and U-Tad Tech Institute.

ACADEMIA PROJECTS

Developed a stochastic reaction-diffusion model to investigate granuloma formation during tuberculosis infection at the alveolar and secondary lobule levels of the pulmonary geometry. Simulated host-pathogen interactions using coupled differential equations for multiple immune cell types and bacillary states. Conducted sensitivity analyses and uncertainty quantification to identify key factors affecting lesion size and encapsulation. **Department of Physics, ETSETB (UPC)** Apr2025–Jun2025

Electrical characterization of metal-semiconductor interfaces using the Transfer Length Method (TLM) across Ag/cSi-DNA samples, extracting contact resistance and sheet resistance. Explored the role of nanometric DNA interlayers in conductivity enhancement. **Group of Characterization of materials - EEBE(UPC)** Mar2025

Experimental study of optical properties in metallic and nanoparticle-enhanced glass samples using UV-Vis spectrophotometry. Measured reflectance of metallic surfaces and extinction spectra of glasses doped nanoparticles. Analyzed light-matter interactions including scattering, absorption, and surface plasmon resonance (SPR), and related spectral features to particle size and composition using Mie theory. Estimated nanoparticle sizes and alloy compositions through Lorentzian peak fitting and optical density modeling. **Group of Characterization of materials, Besòs Campus (UPC)** Feb2025-Mar2025

Design, implementation, and experimental characterization of RF subsystems including VCOs, mixers, amplifiers, and horn antennas for X-band FM-CW Doppler radar applications. Processed baseband signals and explored interferometric radar for precise incremental measurements. **EEF Group, Signal Theory and Communications Dept. (UPC)** Feb2025

Review on Handheld probe integrating laser diode and ultrasound transducer array for ultrasound/photoacoustic dual imaging. **Biomedical Photonics department, ICFO-CD6-UPC** Nov2024-Jan2025

Case Study of Coherence Resonance in a Noise-Driven Excitable System using Hodgkin-Huxley model for neuron system. Studied the influence of the amount of neurons connections on the communication and response to excitations using a hypothetical model with Stochastic Perturbations. **BioPhysics department, ETSETB(UPC)** Dec2024

End-to-end design, implementation, and calibration of a Network Function Measurer for Comprehensive Bode analysis and characterization of systems, including superdiode technology and VCO deployment with Phase-Locked Loop (PLL) integration. Tools: KiCad, LabView, Arduino, STM32. **Physics Engineering department, ETSETB(UPC)** Nov2024-Jan2025

Review on next-generation Quantum Ghost Imaging radar technologies. Development and simulation of qubit systems and algorithms. Tools: Qiskit (Python, IBM). **Quantum Optical Technologies department, ETSETB(UPC)** Nov2024-Dec2024

Case Study of Stochastic Gene Expression in a Single Cell. Biochemical process Noise analysis in gene expression (intrinsic) and fluctuations in other cellular components (extrinsic). Developed ODE Deterministic Computational model and Stochastic Simulations. **BioPhysics department, ETSETB(UPC)** Oct2024

Minimization of Triangle Distortion and Convergence Analysis to enhance the efficiency of PDE finite element analysis resolution, leveraging Multivariate Newton's Method variations for optimized mesh refinement and accuracy.
Numerical Calculus department, FME(UPC) Oct2024-Dec2024

Generation of outfit recommendations based on a given initial product. Fashion matching based on the nuanced patterns and visual details that dictate whether products pair well together, often going beyond mere product metadata. Used Random Forests and other Deep ML techniques. **MANGO Challenge, Datathon FME 2023** Mar2024

Experimental Dosimetry Characterization for Radiation Oncology. Comprehensive analysis of the operational mechanisms of the accelerators utilized in this therapeutic modality and the procedures used to design and measure the radiation dose administered to the patient. **Oncologic Institute TEKNON, Quirónsalud Group** Mar2024-May2024

AI agent development to competite in a structured game setting. Path-finding algorithm development, game theory and ML model design, training, and deployment for real-time AI decision-making and optimization. Worked with reinforcement learning and heuristic-based approaches to improve adaptability and performance. Tools: Java, JetBrains IntelliJ.
ColisseumAI Challenge 2023 Sep2023

Review on theoretical and experimental aspects of Bose-Einstein Condensates, focusing on properties, cooling methods, and applications. Analyzed quantum dynamics using the Gross-Pitaevskii equation and studied techniques like laser and evaporative cooling. Investigated BEC applications in quantum computing, superfluidity, and precision measurement.
Quantum Physics Department, ETSETB(UPC) Oct2022-Dec2022

LANGUAGE SKILLS

Spanish *NATIVE*
Catalan/Valencian: *Certificat Superior JQCV, C2*
English: *Advanced Cambridge ESOL Certificate, C1* *ADVANCED FLUENT*

MORE EDUCATION. OTHER ACTIVITIES AND FONDNESS

Feb 2022 **Persuasive communication** and relational styles course - 5 fars Institute
Marketing, interpersonal, team and public communication for business and science.

2020 **Data Analytics and Assessment for business** - Imperial Collage of London
Finance and economics course with Microsoft tools.

2013-2022 **Violin Professional Studies** - Conservatori Mestre Vert, Carcaixent, Valencia.
Professional training in violin (+10 years). Harmonic, analytical, piano and orchestral knowledge.

ATHLETICS AND VOLUNTEERING

2007-Current **Handball:** *Regional (Valencia) top3 & National (Spain) Top16 u14 team with C.B. Maristas Algemesí.*
2022 Regional (Catalonia) top1 team achieving league promotion with C.H. Esplugues
2023 National (Spain) top4 u23 university team as regional's (Catalonia) top1 university.
Trainer assistant & mentor of young players, helping to improve technical skills, teamwork & game strategy.

2016-2021 **Youth Organizations volunteer** - Juniors Sant Jaume, MAux joves, Algemesí.
Weekly activities aimed at teenagers, focusing on social development and ethical awareness.

2015-2019 **Mathematical Talent Stimulation (ESTALMAT CV)**
Advanced mathematics contest for young students. Contact with physics and mathematics eminences. Valencian universities collaboration.

OTHER SKILLS & IMPLEMENTS

Programming Languages	Python, C++, C, MATLAB, AMPL, R, Java, SQL, JavaScript, Maple
Python Packages	Numpy, Scipy, Matplotlib, PyTorch, TensorFlow, Pandas, Manim
ML/HPC	XGBoost, LightGBM, Numba, Dask, Ray, Keras, Horovod, Modin
Software & Tools	Git, LaTeX, VIM, Excel, Mathematica Spyder, AutoCad, SolidWorks,
Professional tools	AWS, JIRA, Confluence, Slack, Linear, GitLab, Zapier
Electronics	KiCAD, Proteus, Eagle, STM32, Arduino, Raspberry
Music	DaVinci, Musescore